Submission Declarations — Electrical Engineering (Archiv für Elektrotechnik, Springer Nature)

1. Manuscript Details

• Title: PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Structural Recovery for Resilient Power Transformer Thermal Dynamics Forecasting

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2. Availability of Data and Materials

The ETTh1 and ETTh2 benchmark datasets utilized in this study are standardized open-source datasets from the Electricity Transformer Temperature (ETT) benchmark (Zhou et al., AAAI 2021), freely available at: https://github.com/zhouhaoyi/ETDataset. No restricted, private, or commercial telemetry was used.

3. Code Availability

• Repository URL: https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting (MIT License)
Complete PyTorch source code (`pso_tcwavkan_v91.py`), benchmark evaluation matrices (`TABLE1_BENCHMARK_FINAL.csv`), ablation data (`TABLE2_ABLATION_FINAL.csv`), noise robustness evaluations (`TABLE5_NOISE_ROBUSTNESS_FINAL.csv`), and statistical audit scripts are publicly accessible in the repository. Extended multi-seed checkpoints and logs are archived on Kaggle Cloud and available upon reasonable request.

4. Ethical & Financial Declarations

• Competing Interests: The author declares that he has no financial, personal, or professional competing interests that could have appeared to influence the work reported in this paper.
• Funding: The author received no external funding from any public, commercial, or non-profit funding agency for this research. Computational GPU resources were provided via Kaggle Cloud environments at zero cost.
• Author's Contributions: Nguyen Phu Toan is the sole author of this manuscript. The author conceived the mathematical formulation, proved Proposition 1, implemented the codebase, conducted all empirical experiments, analyzed the benchmark data, and drafted and finalized the manuscript.
• Ethics Approval & Consent to Participate: Not applicable. This study involves only publicly available open-source sensor datasets and does not involve human participants, animal subjects, or identifiable personal data.
• Consent for Publication: Not applicable. This manuscript contains no individual person's personal identifying data, images, or details in any form.